

## ST JOSEPH, MI, USA

WMO: 998012

Lat: 42.0980N

Lon: 86.4940W

Elev: 184

StdP: 99.13

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -14.9      | -12.2 | N/A                           | N/A | N/A  | N/A | N/A | N/A  | 14.3                  | -2.5 | 13.0 | -3.1 | 4.1                   | 160  | 0.477 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |     |      |     |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|-----|------|-----|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%  |      | 2%  |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB  | MCDB | WB  | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k) | (l)  | (m) | (n)  | (o)                  | (p)  |
| 7             | 6.8                    | 29.9            | N/A  | 28.3 | N/A  | 27.0 | N/A  | N/A                 | N/A  | N/A | N/A  | N/A | N/A  | 2.6                  | 180  |

| Dehumidification DP/MCDB and HR |     |      |     |     |      |     |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-----|------|-----|-----|------|-----|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |     |      | 1%  |     |      | 2%  |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR  | MCDB | DP  | HR  | MCDB | DP  | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b) | (c)  | (d) | (e) | (f)  | (g) | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| N/A                             | N/A | N/A  | N/A | N/A | N/A  | N/A | N/A | N/A  | N/A           | N/A  | N/A  | N/A  | N/A  | N/A  | N/A               |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |  |  |
|-------------------|------|-----|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|--|--|
|                   |      |     | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |  |  |
| 1%                | 2.5% | 5%  | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max |  |  |
| (a)               | (b)  | (c) | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |  |  |
| 12.7              | 11.1 | 9.5 | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |
|                   |      |     | DB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |
|                   |      |     | WB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 10.3   | -3.4  | -2.5  | 2.5   | 9.2   | 14.9  | 19.7  | 22.6  | 21.8  | 18.4  | 12.3  | 5.9   | 1.1   |
|                                                                                  | DBStd       | 10.04  | 5.63  | 5.18  | 5.28  | 4.45  | 4.75  | 3.56  | 3.24  | 2.65  | 3.68  | 3.97  | 4.83  | 4.66  |
|                                                                                  | HDD10.0     | 1528   | 417   | 352   | 242   | 68    | 9     | 0     | 0     | 0     | 0     | 20    | 141   | 279   |
|                                                                                  | HDD18.3     | 3337   | 675   | 584   | 491   | 277   | 130   | 27    | 5     | 4     | 44    | 192   | 374   | 534   |
|                                                                                  | CDD10.0     | 1623   | 1     | 0     | 11    | 43    | 161   | 291   | 391   | 364   | 252   | 91    | 17    | 3     |
|                                                                                  | CDD18.3     | 390    | 0     | 0     | 1     | 2     | 22    | 67    | 137   | 111   | 46    | 4     | 0     | 0     |
|                                                                                  | CDH23.3     | 1924   | 0     | 0     | 2     | 14    | 116   | 316   | 794   | 486   | 184   | 13    | 0     | 0     |
|                                                                                  | CDH26.7     | 351    | 0     | 0     | 0     | 1     | 23    | 55    | 173   | 71    | 27    | 1     | 0     | 0     |
| Wind                                                                             | WSAvg       | 3.6    | 4.9   | 4.2   | 3.6   | 3.6   | 2.7   | 2.6   | 2.7   | 2.5   | 2.8   | 3.7   | 4.7   | 4.6   |
| Precipitation                                                                    | PrecAvg     | 894    | 59    | 46    | 50    | 75    | 86    | 89    | 84    | 102   | 89    | 92    | 80    | 58    |
|                                                                                  | PrecMax     | 1149   | 125   | 110   | 106   | 132   | 169   | 194   | 182   | 277   | 280   | 221   | 184   | 144   |
|                                                                                  | PrecMin     | 696    | 15    | 13    | 13    | 8     | 18    | 26    | 20    | 37    | 22    | 12    | 28    | 25    |
|                                                                                  | PrecStd     | 130    | 29    | 23    | 26    | 34    | 44    | 47    | 38    | 56    | 59    | 53    | 38    | 28    |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 14.4  | 13.8  | 23.2  | 25.4  | 29.5  | 30.4  | 32.3  | 30.7  | 29.5  | 25.2  | 19.7  | 16.3  |
|                                                                                  |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  | 2%          | DB     | 10.4  | 10.1  | 18.7  | 22.4  | 26.3  | 28.3  | 30.3  | 28.6  | 27.2  | 22.0  | 17.0  | 12.9  |
|                                                                                  |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  | 5%          | DB     | 6.3   | 6.5   | 14.3  | 19.3  | 24.1  | 26.9  | 28.5  | 27.0  | 25.5  | 20.4  | 14.9  | 9.8   |
|                                                                                  |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  | 10%         | DB     | 4.0   | 3.8   | 10.3  | 16.6  | 21.9  | 25.3  | 27.1  | 25.8  | 24.0  | 18.5  | 12.8  | 7.5   |
|                                                                                  |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  | 2%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  | 5%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  | 10%         | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 6.2   | 6.3   | 6.7   | 8.1   | 8.3   | 7.4   | 6.8   | 6.9   | 7.4   | 6.9   | 6.5   | 5.5   |
|                                                                                  |             | MCDBR  | 8.0   | 9.9   | 11.6  | 13.1  | 10.7  | 9.6   | 8.7   | 8.6   | 8.8   | 9.1   | 9.0   | 8.5   |
|                                                                                  |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  | 5% WB       | MCDBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
|                                                                                  |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        |        | 0.304 | 0.326 | 0.346 | 0.392 | 0.421 | 0.432 | 0.439 | 0.436 | 0.409 | 0.358 | 0.322 | 0.305 |
|                                                                                  | taud        |        | 2.433 | 2.359 | 2.371 | 2.277 | 2.257 | 2.278 | 2.258 | 2.266 | 2.304 | 2.444 | 2.492 | 2.475 |
|                                                                                  | Ebn at Noon |        | 842   | 874   | 899   | 877   | 856   | 846   | 834   | 825   | 820   | 829   | 815   | 809   |
|                                                                                  | Edh at Noon |        | 77    | 97    | 108   | 128   | 134   | 132   | 133   | 128   | 115   | 89    | 73    | 69    |
| All-Sky Solar<br>Radiation                                                       | RadAvg      |        | 1.32  | 2.12  | 3.55  | 4.74  | 5.44  | 6.20  | 6.29  | 5.41  | 4.37  | 2.78  | 1.65  | 1.11  |
|                                                                                  | RadStd      |        | 0.14  | 0.28  | 0.28  | 0.40  | 0.49  | 0.40  | 0.29  | 0.31  | 0.36  | 0.31  | 0.25  | 0.14  |

## Historical Trends

|                         |     | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|-----|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |     |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (d)                     | (e) | (f)   | (g)     | (h)    | (i)     | (j)   | (k)   | (l)         | (m)     |         |         |
| Station Only            |     | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| Regional (58 neighbors) |     | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page